

Comment on “ENN's roadmap for proton-boron fusion based on spherical torus” [Phys. Plasmas **31**, 062507 (2024)]

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This comment discusses the feasibility of hot ion mode $T_i/T_e = 4$ for proton–boron fusion, which is critical for the roadmap proposed in Liu *et al.* [Phys. Plasmas **31**, 062507 (2024)]. The hot ion mode $T_i/T_e = 4$ has been calculated to be far from accessible ($T_i/T_e < 1.5$ for $T_i = 150$ keV) under the most optimal conditions if fusion provides the heating [Xie, *Introduction to Fusion Ignition Principles: Zeroth Order Factors of Fusion Energy Research* (USTC Press, Hefei, 2023)], i.e., that all fusion power serves to heat the ions and that electrons acquire energy only through interactions with ions. We also discuss if hot ion mode of $T_i/T_e = 4$ could be achieved by an ideal heating method, which is much more efficient than fusion itself (near 20 times fusion power for $T_i = 150$ keV) and only heats the ions, whether it makes sense economically.

Topics: [Nuclear energy](#), [Fusion energy](#), [Plasma confinement](#), [Tokamaks](#)

Nuclear fusion with advanced fuels has always been conjectured to be the final resolution for commercial fusion, among which proton–boron fusion is attracting growing attention in recent years. Operating in hot ion mode is one of the proposed approaches to commercial proton–boron fusion.

As was discussed in Ref. [1](#), hot ion mode with high temperature difference between electrons and ions of proton–boron fusion is hard to maintain to the extent that even if all fusion power is converted to heat ions and electrons only acquire energy through their interactions with ions, for $T_i = 150$ keV, the proposed operating temperature for proton–boron fusion, $T_i/T_e < 1.5$. This ratio decreases as T_i decreases.

Here we just translate the content on page 49 of Ref. [1](#).

“We make the simplest calculation to estimate the electron temperature in the hot ion mode of proton–boron fusion.

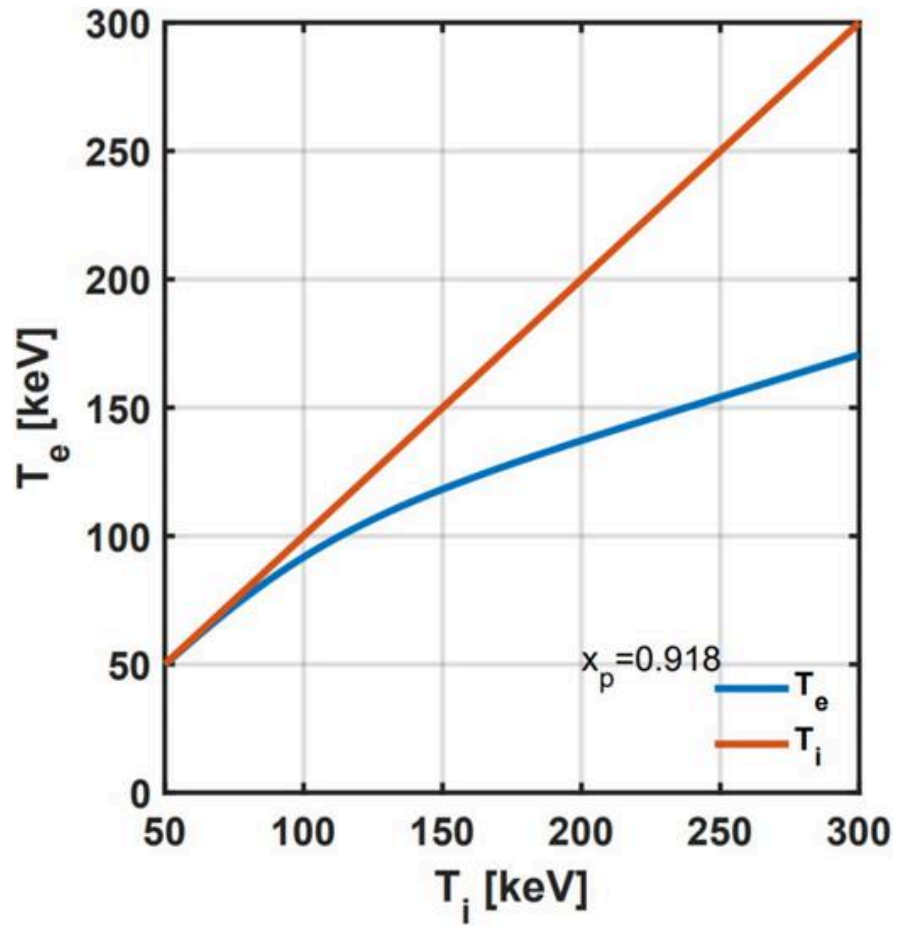
Assume that all the power produced by proton–boron fusion serves to heat ions, and the electrons get energy only through thermal exchange with ions. Neglecting the loss due to radiation and transport, we only do the estimation to the leading

$$-\frac{dW_i}{dt} = \frac{3}{2}k_B \left(\frac{n_p}{\tau_{p,e}} + \frac{n_B}{\tau_{B,e}} \right) (T_i - T_e) = f_{ion} P_{fus} \quad (1)$$

order [which is (3.62) in Ref. [1](#). P_{fus} is the fusion power, n_p and n_B are the densities of proton and boron ions respectively, $f_{ion} = Y_+/Y$ is the proportion of ions in all energy carrying particles, which is assumed to be 1 in Ref. [1](#), and Y is the energy emitted by a single nuclear reaction, Y_+ is the energy of charged particles in a single nuclear reaction, and T_i and T_e are the temperatures of ions and electrons, respectively] where $\tau_{p,e}$ and $\tau_{B,e}$ are the thermal exchange times of proton and boron ions, respectively, which can be calculated by Eq. [\(2\)](#) [Eq. (3.48) in Ref. [1](#)], and the result is depicted in [Fig. 1](#). The electron temperature thus calculated is the minimum possible steady state temperature. Taking other effects into consideration, the practical electron temperature will be larger. It can be seen that for $T_i = 300$ keV, $T_e = 170$ keV, indicating $T_e/T_i = 0.57 > 0.5$. For smaller T_i , the difference between T_i and T_e is even smaller. This demonstrates that it is very hard to maintain hot ion mode with large temperature difference between T_i and T_e . This conclusion is similar to Refs. [2](#) and [3](#)."

$$\tau_{ij} = \frac{3\sqrt{2}\pi^{3/2}\epsilon_0^2 m_i m_j}{n_j e^4 Z_i^2 Z_j^2 \ln \Lambda} \left(\frac{k_B T_i}{m_i} + \frac{k_B T_j}{m_j} \right)^{3/2}. \quad (2)$$

FIG. 1.



The relationship between T_i and T_e for hot ion mode of proton–boron fusion. Under the most optimal condition, from Fig. 3.17(a) in Ref. [1](#).

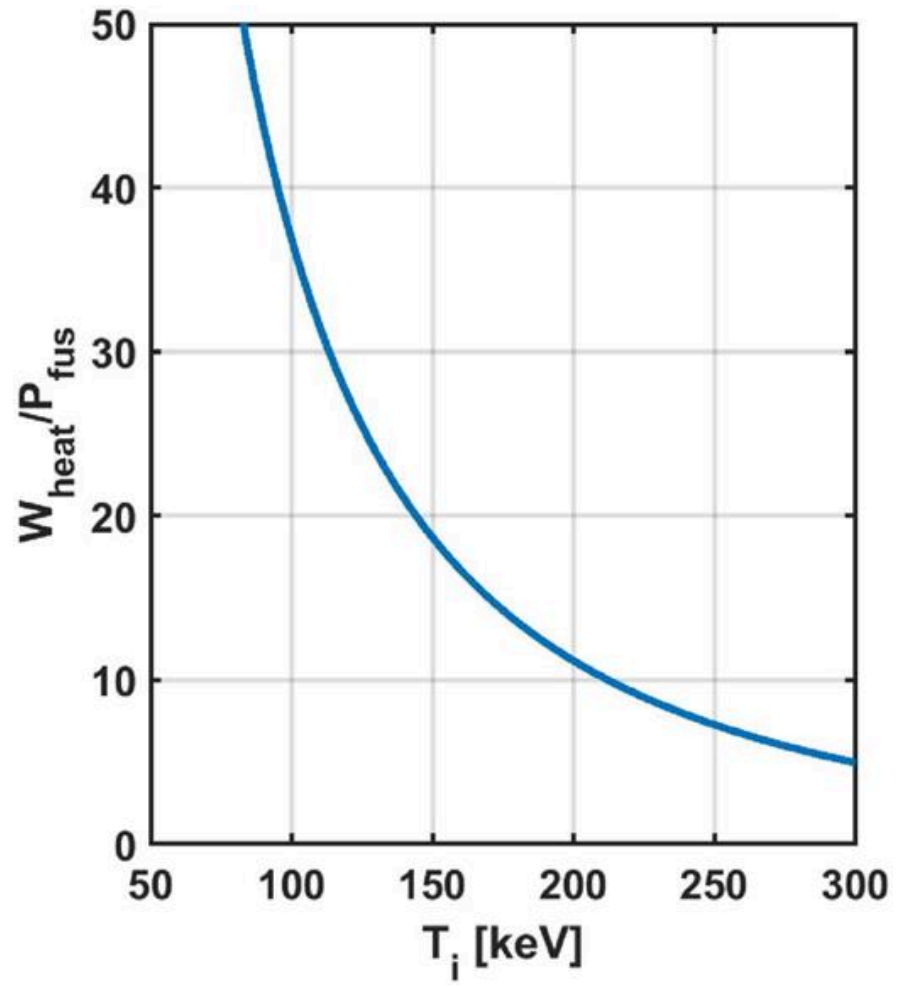
Now let us assume that there is an ideal heating method, to the extreme that it only heats ion, and that it leads to no additional energy loss. This idealized heating enables hot ion mode of $T_i/T_e = 4$. Denoting it by W_{heat} , the balance equation [\(1\)](#) transforms into

$$-\frac{dW_i}{dt} = \frac{3}{2}k_B \left(\frac{n_p}{\tau_{p,e}} + \frac{n_B}{\tau_{B,e}} \right) (T_i - T_e) = f_{ion}P_{fus} + W_{heat}. \quad (3)$$

Equation [\(3\)](#)

actually becomes a linear function for W_{heat} . Adapting the code from <http://hsxie.me/fusionbook/>, we can easily calculate W_{heat} and compare it with the fusion power $f_{ion}P_{fus}$. The result is plotted in [Fig. 2](#).

FIG. 2.

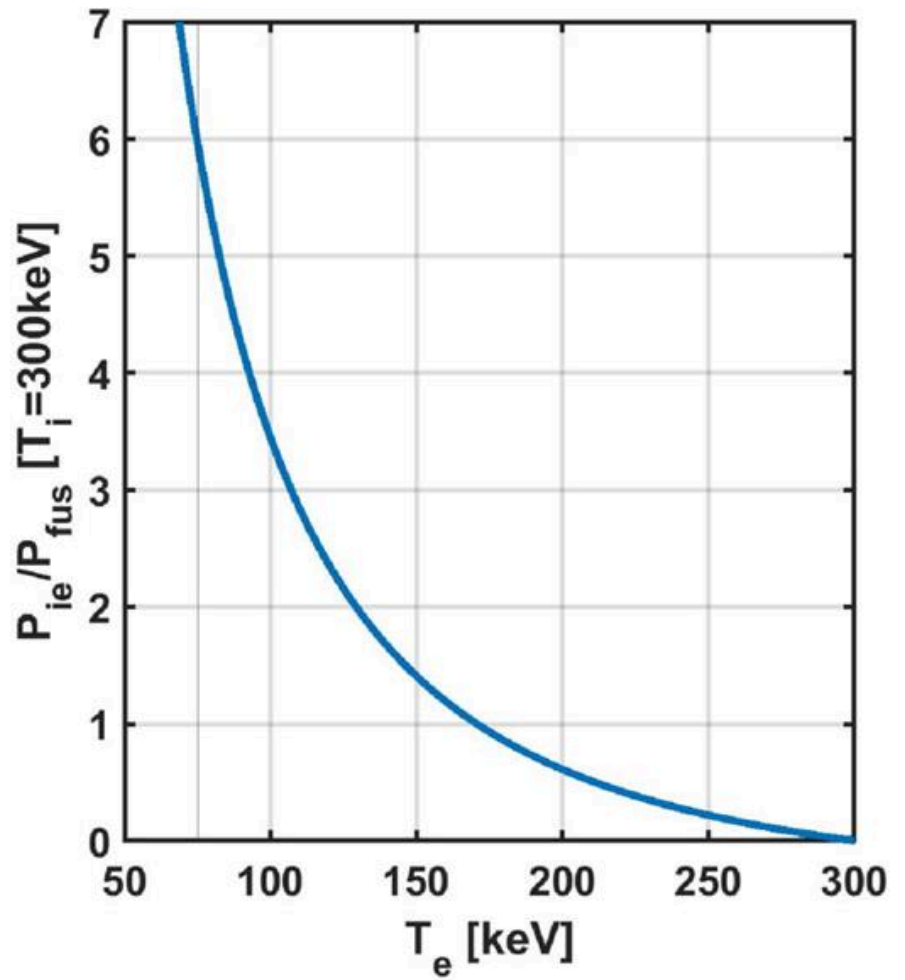


The ratio between heating power and fusion power for $T_i/T_e = 4$.

It is apparent that even the idealized W_{heat} that enables $T_i/T_e = 4$ is way larger than fusion power.

In fact, after slight extension of the axis in Fig. 3.17(b) of Ref. [1](#), i.e., 3, it is already clear that for $T_i = 300$ keV, hot ion mode of $T_i/T_e = 4$ demands a heating that is about five times the fusion power ([Fig. 3](#)).

FIG. 3.



The power of energy exchange between electrons and ions vs fusion power for $T_i = 300$ keV. Adapted from Fig. 3.17(b) of Ref. [1](#).

The situation is, therefore, very awkward: if electron temperature is high, Bremsstrahlung radiation is significant; and if the electron temperature is kept to be low, the power to sustain hot ion mode would be much larger than fusion power. Now imagine that we have finally succeeded in finding such an ideal heating method, a question emerges naturally: Why do we not just use its power than bother to do fusion?

This analysis also leads to a question on the validity of the scaling law for the energy confinement time τ_E of proton–boron fusion.^{[4](#)} Were it rigorous, the fusion energy gain factor Q calculated from the triproduct $nT\tau_E$ or the power balance with term $\frac{W_{th}}{\tau_E}$ would be large.^{[4](#)} However if we calculate Q directly, then it is vastly smaller. If we were not lucky enough to find such an ideal heating, τ_E does not exist at all for proton–boron fusion with $T_i/T_e = 4$.

We thank Dr. Xie for the permission to reproduce the figures in his book.

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